# Anand Bhattad


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## EDUCATION

**University of Illinois, Urbana-Champaign (UIUC)**, USA

Ph.D., Computer Science — *Jan'19 - Aug 2023*
*Advisor:* David Forsyth
*Thesis committee:* David Forsyth (chair), Alyosha Efros, Bill Freeman, Derek Hoiem, Shenlong Wang and Svetlana Lazebnik

M.S., Computer Science — *Aug'17 - Dec'18*
M.S., Civil Engineering — *Aug'15 - Aug'17*

**National Institute of Technology Karnataka (NITK) Surathkal**, India

B.Tech., Civil Engineering — *July'11 - May'15*

## RESEARCH EXPERIENCE

**Toyota Technological Institute at Chicago**, Chicago — *Sept'23 - present*
*Research Assistant Professor*

**Allen Institute of Artificial Intelligence**, Seattle/Champaign(remote) — *Feb'23 - April'23*
*Research Intern* with *Ani Kembhavi*

**Intel ISL**, Santa Clara/Champaign (remote) — *Feb'21 - Aug'21*
*Research Intern* with *Stephan R. Richter.*

**NVIDIA ADLR**, Santa Clara/Champaign (remote) — *May'20 - Dec'20*
*Research Intern* with *Andrew Tao and Bryan Catanzaro*

**Fyusion Inc**, San Francisco — *May'19 - Aug'19*
*Research Intern* with *Abhishek Kar and Rodrigo Ortiz Cayon*

**Siemens Corporate Technology**, Princeton — *May'17 - Aug'17*
*Computer Vision & DL Intern* with *Jan Ernst and Ziyan Wu*

**Computational Sensor Lab**, Michigan State University — *May'14 - July'14*
*Research Intern* with *Prof. Nizar Lajnef and Prof. Amir H. Alavi*

**Earthquake Engineering Research Center**, IIIT Hyderabad — *Jun'13 - July'13*
*Research Intern* with *Prof. Ramacharla Pradeep Kumar*

## PREPRINTS /UNDER REVIEW

7. **Generative Models: What do they know? Do they know things? Let's find out!**
   X. Du, N. Kolkin, G. Shakhnarovich, **A. Bhattad**
   In Submission
6. ***Shadows Don't Lie and Lines Can't Bend!* Generative Models don't know Projective Geometry...for now**
   A. Sarkar, H. Mai, A. Mahapatra, S. Lazebnik, D.A. Forsyth, **A. Bhattad**
   In Submission
5. **UrbanIR: Large-Scale Urban Scene Inverse Rendering from a Single Video**
   Z. Lin, B. Liu, Y. Chen, D.A. Forsyth, J. Huang, **A. Bhattad**, S. Wang
   In Submission
4. **Blocks2World: Controlling Realistic Scenes with Editable Primitives**
   V. Vavilala, S. Jain, R. Vasanth, **A. Bhattad**, D.A. Forsyth
   In Submission

3. **StyLitGAN: Image-based Relighting via Latent Control**
   **A. Bhattad**, J. Soole, D. A. Forsyth
   In Submission
2. **Make It So: Steering StyleGAN for Any Image Inversion and Editing**
   **A. Bhattad**, V. Shah, D. Hoiem, D. A. Forsyth
   In Submission
1. **SIRfyN: Single Image Relighting from your Neighbors**
   D. A. Forsyth, **A. Bhattad**, P. Asthana, Y. Zhong, Y. Wang
   Technical Report

## Conference Papers

8. **StyleGAN Knows Normals, Depth, Albedo, and More**
   **A. Bhattad, D. McKee, D. Hoiem, D. A. Forsyth**
   NeurIPS 2023
7. **OBJect 3DIT: Language-guided 3D-aware Image Editing**
   O. Michel, **A. Bhattad**, E. VanderBilt, R. Krishna, A. Kembhavi, T. Gupta
   NeurIPS 2023
6. **Learning Equivariant Dense Prediction Networks**
   Yuanyi Zhong, **A. Bhattad**, Y. Wang, D. A. Forsyth
   ICCV 2023
5. **Cut-and-Paste Object Insertion by Enabling Deep Image Prior for Reshading**
   **A. Bhattad**, D. A. Forsyth
   3DV 2022
4. **DIVeR: Real-time and Accurate Neural Radiance Fields with Deterministic Integration for Volume Rendering (Best Paper Finalist)**
   L. Wu , J. Y. Lee, **A. Bhattad**, Y. Wang, D. A. Forsyth
   CVPR 2022
3. **View Generalization for Single Image Textured 3D Models**
   **A. Bhattad**, A. Dundar, G. Liu, A. Tao, B. Catanzaro
   CVPR 2021
2. **Unrestricted Adversarial Perturbations via Semantic Manipulation**
   **A. Bhattad***, M. J. Chong*, K. Liang, B. Li, D. A. Forsyth (* for equal contribution)
   ICLR 2020
1. **Improved Style Transfer with Calibrated Metrics**
   M. Yeh*, S. Tang*, **A. Bhattad**, C. Zou, D. A. Forsyth (* for equal contribution)
   WACV 2020

## Workshop Papers

2. **Big but Imperceptible Adversarial Perturbations via Semantic Manipulation**
   **A. Bhattad***, M. J. Chong*, K. Liang, B. Li, D. A. Forsyth (* for equal contribution)
   Computer Vision and Pattern Recognition (CVPR) Workshop 2019, ***Oral Presentation***
1. **Detecting Anomalous Faces with "No Peeking" Autoencoders**
   **A. Bhattad**, J. Rock, D. A. Forsyth
   Computer Vision and Pattern Recognition (CVPR) Workshop, 2018, ***Oral Presentation***

## Students Mentoring

- Xiaodan Du (PhD; present) | Project on Generative Models Knowledge
- Ayush Sarkar (PhD; present) | Project on Evaluating Generated Images
- Hanlin Mai (PhD; present) | Project on Evaluating Generated Images
- Amitabh Mahapatra (Undergrad; present) | Project on Evaluating Generated Images
- James Soole (MS; present) | Project on Illumination Physics in Dense Prediction Models
- Feiran Wang (MEngg; Spring 2023) | Project on NeRF + Illumination Physics
- Liwen Wu (BS-MS; 2021-2022) | Paper at CVPR 2022 | Next position: PhD Student at UCSD
- Brian Chen (BS; 2020-2022) | Paper under review | REU at CMU's Robotics Institute
- Kexuan (Klaus) Zou (MS; 2019-2020) | Next position: Software Engineer at NVIDIA
- Anchu Zhu (BS; 2018-2019) | Next position: MS in CS at USC

## Teaching Experience

**Computer Science**, UIUC; *Graduate Teaching Assistant* *Jan'16 - Aug'23*

- Applied Machine Learning (CS498 AML) in Fall'18 | Designed Kaggle competitions
- Data Structures (CS 225) in Spring'17 | Developed Assignments, Exams & Labs
- Introduction to Computing (CS 101) in Spring'16 & Fall'16 | **Rated as Excellent TA**

## Services & Leadership

**Organizer**, CVPR-W on "Big Models Vs Scholars: How Should Academics Adapt?" June 2023

- Co-organizing a CVPR workshop to discuss the potential impact of large vision models on the Academic CV community
- Aims to bring fresh perspectives on redesigning collaborations, access to compute, and curricula.
- Positively shape a more inclusive, equitable, and diversity-rich future for computer vision via discussions and opinions through assorted panels and speaker talks

**Organizer**, External Speaker Series and Vision Lunch Jan 2021-Present

- Hosting the first external speaker series, inviting researchers outside of UIUC for technical talks
- Organizing computer vision reading and presentation group in the CS department of UIUC

**Reviewer**, Computer Vision Conferences Nov 2018-Present

- Serving as reviewer for CVPR, NeurIPS, ECCV and ICCV conferences and workshops.
- **Outstanding Emergency Reviewer** for CVPR 2021
- **Outstanding Reviewer** for ICCV 2023

**Program Committee** Jun 2019

- Adversarial Machine Learning in Real-World Computer Vision Systems, CVPR 2019 Workshop
- Security and Privacy in Machine Learning, ICML 2019 Workshop

**Organizer**, Vision Coffee, Urbana-Champaign *May'18 - Aug'18*

- Vision and Deep Learning paper/topic weekly discussion group
- Finalized agenda/topic, location and initiated discussions

**Co-Founder**, SchoolEngg *Apr'13 - May'15*

- Forum for high-school students; providing the firsthand knowledge of all engineering disciplines
- Currently, renamed as *PrepLift*, an education counselling start-up in India